

202401465_Lab_3_Report

Prompts used :

Case Study :

```
case_study = """
    Smart Campus Cafeteria / Food Court Ordering & Feedback System
    Most college campuses run their cafeterias manually: students wait in unpredictably long
    queues, place orders verbally, and wait again without knowing prep time. This causes overcrow
    ding and frustration. Cash/card handling is slow, and manual errors are common.
    Vendors manage inventory manually, leading to food wastage or stock-outs. Feedback is inf
    ormal (word of mouth).
    Administration wants to oversee pricing, hygiene, and value without relying solely on man
    ual inspections. Hostel wardens also track meal plans for hostel students.
    The new system should allow digital ordering, real-time tracking, inventory/sales managem
    ent for vendors, and a structured feedback loop. It should also give admin and wardens oversi
    ght tools.
    """
```

Stakeholder Identification Prompt :

```
Analyze the following case study and identify all primary, secondary, and administrative stak
eholders.
Provide a structured list of roles with a brief 1-sentence justification for each.

Case Study:
{case_study}
```

Stakeholder Goals Identification Prompt:

Output : goals (used in next prompt)

```
Based on the case study and the identified stakeholders, map out the explicit pain points and
desired outcomes (goals) for each stakeholder group.

Case Study: {case_study}
Stakeholders: {stakeholders}
Provide a structured breakdown for each stakeholder group.
```

Elicitation Technique Selection Prompt :

Output : techniques (used in next prompt)

```
Based on the stakeholders and their goals, select the optimal requirements elicitation techni
que for each group (e.g., Questionnaires/Surveys, Semi-structured Interviews, Audits, Observa
tions).
Justify your pairing based on group size, accessibility, and depth of needed insights.
```

Stakeholders & Goals:
{goals}

Elicitation Execution Prompt :

Output : instruments (used in next prompt)

Execute the selected elicitation techniques by generating the actual instruments.
For example, **if** surveys were chosen **for** students, draft **4-5** key survey questions. If interviews were chosen **for** cafeteria vendors, draft **4-5** specific interview questions. Include administrative oversight questions **if** applicable.

Techniques Selected:

{techniques}

FR and NFR Generation Prompt :

Output : requirements (final output to showcase)

Synthesize **all** the gathered data, **case** study, **and** elicitation instruments to **compile** the final system requirements.

Output two structured lists:

- 1.** Functional Requirements (FRs): Core system features **and** behaviors (e.g., User Auth, Digital Menu, Tracking).
- 2.** Non-Functional Requirements (NFRs): Quality attributes (e.g., Peak load handling, Data security, UI usability).

Case Study: {case_study}

Elicitation Data: {instruments}

The Chosen Feedback Mechanism: "Human-in-the-Loop (HITL) with Iterative Refinement

- **Execution Pause:** The pipeline stops automatically after generating the output for each of the 5 stages.
- **Terminal Prompt:** It displays the AI's output and asks the user to either approve it or provide text feedback.
- **Approval Gate:** Pressing **Enter** (leaving it blank) approves the output and pushes it to the next stage.
- **Iterative Refinement:** If the user types feedback (e.g., "Add IT admins"), a secondary prompt feeds this correction back to the AI. The AI revises its answer, and the loop repeats until the user approves.

Reasoning: Why this specific mechanism?

- **Quality Control:** Elicitation requires precision. If an error happens in Step 1, it will pollute all following steps. Forcing human approval catches errors early.
- **Practicality over Complexity:** A manual terminal review is highly reliable. Having the "LLM judge itself" is risky and often leads to circular logic and hallucinations.
- **True Steering:** It fulfills the lab requirement to actually "steer" the output, rather than just accepting or rejecting it.

Notable Prompting & Design Decisions

- **Context Forwarding:** The output of Step 1 (Stakeholders) is explicitly passed as a variable into the prompt for Step 2 (Goals). This forces the LLM to stay consistent and not hallucinate new groups.

- **String Output Parsers:** Used LangChain's `StrOutputParser()` at the end of every chain. This ensures the pipeline passes clean raw text between stages instead of complex, unreadable internal AI objects.
- **Strict Refinement Prompts:** The refinement prompt explicitly orders the AI to revise the text *"based strictly on the feedback provided."* This stops the AI from completely rewriting a good answer just to fix a tiny typo